

Windows Security Monitoring Investigation Report

Prepared by

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SOC Analyst Home Lab Project

Tools Used

- Splunk Enterprise
- Windows 11
- Windows Security Logs
- Splunk Universal Forwarder
- VirtualBox

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1. Project Overview

This project demonstrates the implementation of a **Windows Security Monitoring solution** using **Splunk Enterprise** in a controlled Windows 11 home lab environment. The objective was to collect, monitor, analyze, and investigate Windows Security Event Logs that are commonly reviewed by Security Operations Center (SOC) analysts during security monitoring and incident response.

Windows Security Event Logs were forwarded from the Windows 11 virtual machine to Splunk Enterprise using the Splunk Universal Forwarder. The investigation focused on authentication events, privileged logons, account management activities, and authentication trends using Windows Security Event IDs.

Several Search Processing Language (SPL) queries were developed to identify successful logins, failed authentication attempts, privileged account usage, and new user account creation. These events were correlated to simulate the investigation workflow followed by SOC analysts while validating authentication-related security alerts.

The project also included the development of an interactive SOC dashboard displaying authentication statistics, login trends, and frequently used accounts to improve security visibility and accelerate incident investigation.

The investigation provided practical experience in Windows Security monitoring, SIEM operations, event correlation, authentication analysis, and MITRE ATT&CK mapping while demonstrating real-world SOC investigation techniques.

2. Lab Environment

Component	Details
Operating System	Windows11
SIEM Platform	Splunk Enterprise
Log Collection	Splunk Universal Forwarder
Virtualization	VirtualBox
Log Source	Windows Security Logs
Investigation Tool	Splunk Enterprise

3.Incident Summary

Field	Value
Incident ID	SOC - 002
Incident Type	Windows Authentication Monitoring
Severity	Medium
Status	Closed
Detection Method	Splunk Enterprise SPL Searches
Log Source	Windows Security Event Logs
Analyst	T.Ramakrishna

During routine security monitoring, several Windows Security events were detected and analyzed using Splunk Enterprise. The investigation focused on successful authentication events, failed login attempts, privileged logons, and user account creation events collected from Windows Security Logs. SPL searches were used to correlate authentication activities and validate whether the observed events represented legitimate user behavior or suspicious activity.

4. Investigation Timeline

Time	Activity
9:15 AM	Successful Logon detected (Event ID 4624)
9:18 AM	Failed Login Attempt detected (Event ID 4625)
9:22 AM	Privileged Logon detected (Event ID 4672)
9:25 AM	New User Account detected (Event ID 4720)
9:30 AM	Authentication Timeline Correlation
9:35 AM	Investigation completed
9:38 AM	Incident closed

5. Investigation Process

5.1. Successful Logon Investigation

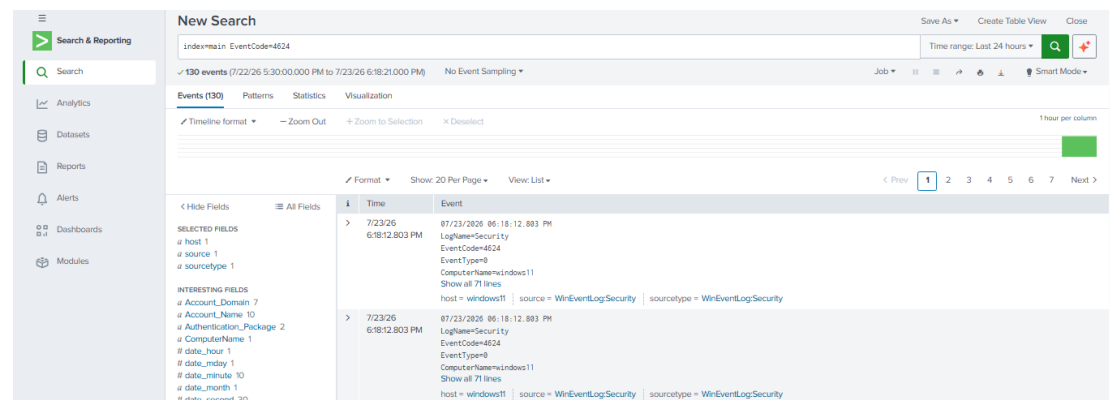
Objective

The investigation began by reviewing successful authentication events to verify normal user login activity and identify unusual authentication behavior.

SPL Query

index=main EventCode=4624

Figure 1: Successful Logon Investigation (4624)



Time	Event
7/23/26 6:18:12.803 PM	LogName=Security EventCode=4624 EventType=8 ComputerName=windows11 Show all 71 lines host = windows11 source = WinEventLogSecurity sourcetype = WinEventLogSecurity
7/23/26 6:18:12.803 PM	LogName=Security EventCode=4624 EventType=8 ComputerName=windows11 Show all 71 lines host = windows11 source = WinEventLogSecurity sourcetype = WinEventLogSecurity

Investigation

Windows Security Event ID 4624 records every successful authentication. The investigation reviewed the username, logon type, workstation, source IP address, and login timestamp to determine whether the authentication originated from an expected source.

Observation

Successful logon events were collected successfully. Authentication details matched expected user activity within the lab environment.

Analyst Conclusion

The successful authentication events were legitimate and served as the baseline for further correlation with failed logon attempts.

5.2 Failed Logon Investigation

Objective

Analyze failed login attempts to identify possible brute-force attacks or unauthorized authentication attempts.

SPL Query

Index=main EventCode=4625

Figure 2: Failed Logon Investigation (4625)

New Search

index=main EventCode=4625

2 events (7/22/26 5:30:00.000 PM to 7/23/26 6:19:55.000 PM) No Event Sampling

Time range: Last 24 hours

Events (2) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a Account_Domain 2
- a Account_Name 2
- a Authentication_Package 1
- a Caller_Process_Name 1
- a ComputerName 1
- # date_hour 1
- # date_inday 1
- # date_minute 1

i	Time	Event
>	7/23/26 6:05:16.529 PM	07/23/2026 06:05:16.529 PM LogName=Security EventCode=4625 EventType=0 ComputerName=windows11 Show all 61 lines host = windows11 source = WinEventLog\Security sourcetype = WinEventLog\Security
>	7/23/26 6:05:16.529 PM	07/23/2026 06:05:16.529 PM LogName=Security EventCode=4625 EventType=0 ComputerName=windows11 Show all 61 lines host = windows11 source = WinEventLog\Security sourcetype = WinEventLog\Security

Investigation

Windows Event ID 4625 records unsuccessful authentication attempts. Multiple failed logins against the same account or from the same source can indicate password guessing or brute-force activity.

Observation

The investigation identified failed authentication events. No abnormal pattern indicating brute-force activity was observed in the lab environment.

Analyst Conclusion

Failed logon monitoring successfully detected authentication failures and provided valuable visibility into login attempts.

5.3.Privileged Logon Investigation

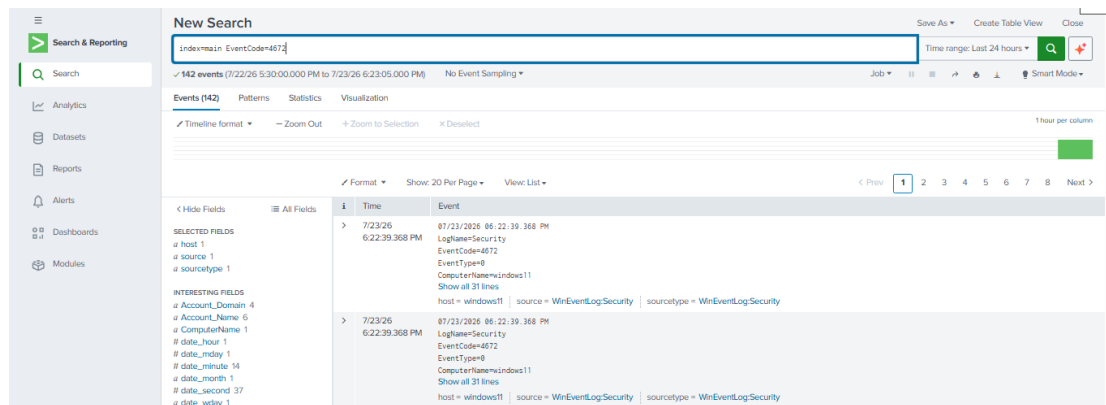
Objective

Identify privileged logons and verify whether administrative privileges were assigned to authorized users.

SPL Query

index=main EventCode=4672

Figure 3: Privileged Logon Investigation (4672)



Investigation

Windows Event ID 4672 is generated whenever special administrative privileges are assigned during logon.

Observation

Privileged logon events were successfully collected and correlated with successful authentication events.

Analyst Conclusion

Administrative activities matched expected behavior within the controlled environment.

5.4 New User Account Investigation

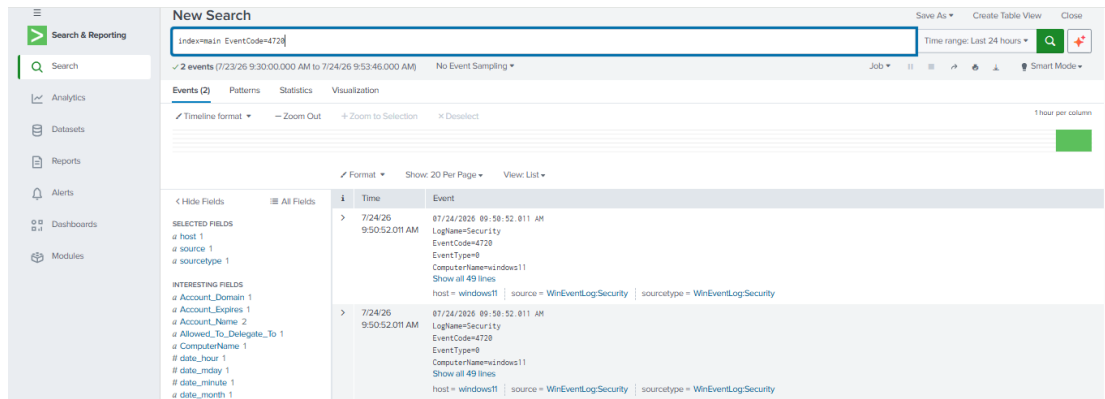
Objective

Monitor newly created user accounts.

SPL Query

index=main EventCode=4720

Figure 4: New User Account Investigation (4720)



Investigation

Event ID 4720 records the creation of a new user account.

Observation

New account creation events were successfully indexed into Splunk.

Analyst Conclusion

Account creation monitoring provides early visibility into unauthorized persistence attempts.

5.5 Authentication Timeline Correlation

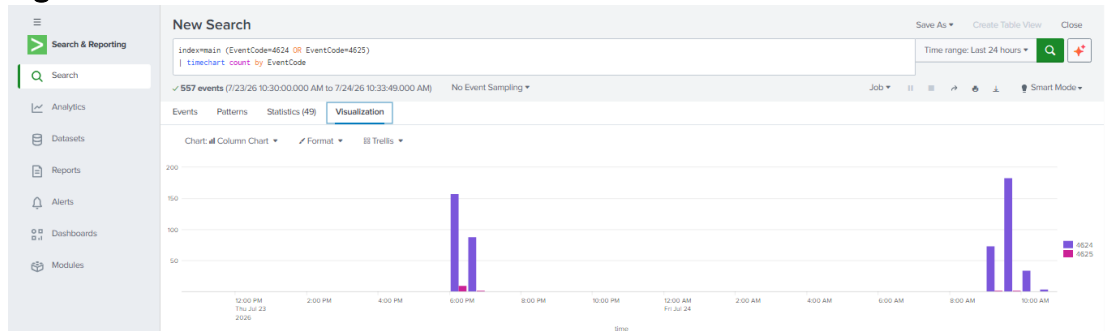
Objective

Correlate authentication events to reconstruct the sequence of activities.

SPL Query

**index=main (EventCode=4624 OR EventCode=4625)
| timechart count by EventCode**

Figure 5: Authentication Timeline



Investigation

Authentication events were correlated chronologically to identify abnormal login spikes or repeated failures.

Observation

The timeline clearly displayed successful and failed authentication activity.

Analyst Conclusion

Authentication correlation significantly improves detection of brute-force attacks and account compromise.

6. SPL Queries Used

The following SPL searches were used during the investigation to identify, analyze, and correlate Windows Security events collected from the Windows Security Event Log in Splunk Enterprise.

Investigation	SPL Query	Purpose
Successful Logons (Event ID 4624)	<code>index=main EventCode=4624</code>	Detect successful user authentication events and review the user account, logon type, workstation, and source network address.
Failed Logons (Event ID 4625)	<code>index=main EventCode=4625</code>	Identify failed authentication attempts that may indicate incorrect passwords, password spraying, or brute-force attacks.
Privileged Logons (Event ID 4672)	<code>index=main EventCode=4672</code>	Detect logons where administrative or special privileges are assigned to a user account.
New User Account Creation (Event ID 4720)	<code>index=main EventCode=4720</code>	Detect newly created local or domain user accounts to identify potential unauthorized account creation.
Authentication Timeline	<code>index=main (EventCode=4624 OR EventCode=4625) timechart count by EventCode</code>	Visualize successful and failed authentication events over time to identify login spikes or brute-force activity.

7. Investigation Findings

- Windows Security logs were successfully forwarded into Splunk Enterprise.
- Successful logons were monitored using Event ID 4624.
- Failed authentication attempts were monitored using Event ID 4625.
- Administrative logons were detected using Event ID 4672.
- New user account creation was detected using Event ID 4720.
- Authentication timelines helped identify login trends.
- Dashboard visualization improved SOC monitoring efficiency.
- Authentication events were successfully correlated.

8. MITRE ATT&CK Mapping

Activity	Technique	Technique ID
Successful Logon	Valid Accounts	T1078
Failed Login Attempts	Brute Force	T1110
Privileged Logon	Account Manipulation / Privilege Escalation	T1098
Account Creation	Create Account	T1136

9. Indicators of Compromise (IOCs)

IOC Type	Description
Event ID	4624
Event ID	4625
Event ID	4672
Event ID	4720
Account Name	Windows 11
Log Source	WinEventLog:Security
Computer	Windows 11

10. Conclusion

This project demonstrated the implementation of centralized Windows Security monitoring using Splunk Enterprise. Authentication events were successfully collected, analyzed, and correlated to investigate user logon activity, failed authentication attempts, privileged access, and account management events. The dashboard and SPL searches provided comprehensive visibility into Windows authentication behavior and strengthened practical skills in SIEM operations, event correlation, and SOC investigation workflows.

11. Recommendations

- Enable alerts for repeated failed logon attempts.
- Monitor privileged account activity continuously.
- Review new account creation daily.
- Implement Multi-Factor Authentication (MFA).
- Integrate Sysmon logs for enhanced endpoint visibility.
- Correlate Windows Security Logs with firewall and EDR telemetry.
- Regularly tune SIEM detection rules to reduce false positives.
- Map detections to the MITRE ATT&CK framework to improve detection coverage.